

MASON CITY, IA, USA

WMO: 725485

Lat: 43.154N

Lon: 93.327W

Elev: 373

StdP: 96.92

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-21.4	-27.6	0.3	-23.7	-24.9	0.4	-21.1	15.4	-7.8	14.1	-10.3	4.8	320	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.3	23.7	30.7	22.9	29.2	22.0	25.6	30.1	24.4	28.6	23.4	27.6	6.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	20.0	28.4	23.1	18.6	27.3	21.9	17.4	26.1	80.8	30.3	75.7	28.5	71.4	27.8	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.5	10.4	DB	-28.2	34.9	3.1	1.4	-30.5	36.0	-32.3	36.8	-34.0	37.6	-36.3	38.7	(4)
(5)			WB	-28.4	27.5	3.1	1.2	-30.6	28.4	-32.4	29.1	-34.1	29.7	-36.4	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-8.9	-6.6	0.5	8.0	14.7	20.4	22.2	20.7	16.8	9.5	1.5	-5.7	(6)	
	DBStd	12.17	7.19	6.99	6.79	5.53	4.73	3.62	3.14	3.07	4.55	5.31	5.92	6.52	(7)	
	HDD10.0	2287	585	464	302	99	12	0	0	0	4	76	261	486	(8)	
	HDD18.3	4181	843	697	552	311	132	20	5	12	81	279	505	744	(9)	
	CDD10.0	1503	0	0	8	41	158	312	379	332	209	59	6	0	(10)	
	CDD18.3	354	0	0	0	2	20	82	126	86	36	4	0	0	(11)	
	CDH23.3	3391	0	0	3	42	231	769	1179	735	376	54	1	0	(12)	
	CDH26.7	1062	0	0	0	11	67	252	405	207	108	12	0	0	(13)	
(14) Wind	WSAvg	4.9	5.6	5.6	5.6	5.9	5.4	4.6	3.7	3.4	4.0	4.9	5.3	5.3	(14)	
(15) Precipitation	PrecAvg	864	22	24	51	82	115	117	118	101	87	59	41	29	(15)	
	PrecMax	1186	62	90	123	208	328	290	310	400	366	143	106	100	(16)	
	PrecMin	435	2	1	10	21	21	11	14	24	13	2	0	5	(17)	
	PrecStd	153	16	16	29	41	55	66	69	73	70	38	32	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	12.1	22.4	28.1	31.8	33.8	34.6	32.8	32.1	28.4	20.4	12.5	(19)	
		MCWB	4.7	8.9	14.3	17.3	20.8	23.5	25.5	24.2	21.9	18.3	13.3	10.0	(20)	
	2%	DB	4.3	7.4	17.5	23.7	28.4	31.5	32.4	30.8	29.6	24.4	16.4	7.8	(21)	
		MCWB	2.3	5.4	12.9	14.8	18.3	22.4	24.9	24.2	21.5	16.5	11.7	5.6	(22)	
	5%	DB	2.1	4.6	13.6	20.5	25.8	29.6	30.6	29.0	27.5	21.2	13.1	4.8	(23)	
		MCWB	0.8	2.9	10.1	12.8	17.3	21.1	23.8	22.9	19.9	15.3	9.1	3.2	(24)	
	10%	DB	0.6	2.4	10.0	17.7	23.4	27.7	28.9	27.3	25.4	18.4	10.2	2.5	(25)	
		MCWB	-0.2	1.1	7.4	11.4	16.1	20.2	22.7	21.8	18.8	13.4	7.1	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	9.2	16.4	18.4	22.6	25.6	27.6	26.6	24.3	20.8	15.0	10.7	(27)	
		MCDB	8.2	11.4	20.9	25.3	29.4	31.3	32.3	30.8	29.2	25.0	18.9	12.8	(28)	
	2%	WB	2.4	5.4	13.4	16.0	20.5	24.0	26.0	25.0	22.8	18.2	12.3	5.9	(29)	
		MCDB	4.0	7.2	16.7	21.7	26.0	28.8	30.3	29.3	27.9	22.1	15.7	7.4	(30)	
	5%	WB	1.0	3.0	10.3	14.1	18.9	22.8	24.9	23.9	21.4	16.2	9.7	3.2	(31)	
		MCDB	1.9	4.5	13.3	19.3	23.6	27.8	29.1	27.6	25.4	20.0	12.7	4.7	(32)	
	10%	WB	-0.2	1.2	7.2	11.9	17.4	21.5	23.7	22.7	20.1	14.1	7.3	1.3	(33)	
		MCDB	0.7	2.3	10.1	16.5	21.8	26.2	27.7	26.1	23.8	18.2	10.1	2.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	9.7	12.1	11.9	11.5	11.5	11.7	13.3	12.1	10.0	8.4	(35)	
		MCDBR	10.2	10.4	15.1	17.6	15.2	13.4	12.7	12.9	15.2	16.2	14.7	11.0	(36)	
	5% WB	MCWBR	8.6	8.5	10.0	9.9	7.6	6.3	6.4	6.9	7.6	9.1	9.8	8.8	(37)	
		MCWBR	9.3	9.8	13.9	15.2	12.6	11.9	11.2	11.5	12.7	12.8	13.3	10.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.271	0.290	0.314	0.365	0.395	0.417	0.427	0.408	0.377	0.331	0.292	0.268	(40)	
		taud	2.417	2.367	2.423	2.318	2.301	2.281	2.239	2.289	2.345	2.465	2.504	2.446	(41)	
	Ebn at Noon		875	911	931	900	879	856	843	847	849	853	845	845	(42)	
		Edh at Noon	75	94	101	122	128	131	135	124	109	85	69	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.82	2.86	3.77	4.67	5.50	6.13	6.35	5.39	4.36	2.87	1.91	1.41	(44)	
	RadStd		0.19	0.27	0.38	0.47	0.41	0.48	0.32	0.37	0.30	0.37	0.16	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page